HAND & UPPER LIMB · NEET-SS / FET / INI-SS

Hand infections

1. General principles

- **Commonest organism: *Staphylococcus aureus*** (**MRSA increasingly common in community-acquired hand infections**), then **beta-haemolytic streptococci**; **mixed/anaerobic in bites, diabetics, IV drug users**.
- **Hand anatomy predisposes to spread:** closed compartments (pulp, sheaths, deep palmar spaces), **lymphatic drainage to the dorsum (dorsal swelling even with palmar infection)**.
- **Principles: elevation, immobilisation in the "safe position" (intrinsic plus: wrist extended about 30 degrees, MCPJs flexed 70-90 degrees, IPJs extended)**, **antibiotics**, **early surgical drainage of pus**, **debridement**, **early mobilisation** once infection controlled; **check tetanus status**; **control diabetes**.
- **Abscess (fluctuant collection) = drainage; cellulitis = antibiotics + elevation** (**reassess within 24-48 hours; failure to improve: suspect abscess**).

2. Paronychia

- **Infection of the nail fold (perionychium)**: **commonest hand infection**; **acute: *S. aureus*** (nail biting, manicure, hangnail).
- **Treatment: early (cellulitis): warm soaks, oral antibiotics**; **abscess: incision/elevation of the nail fold (blade directed away from the nail bed), partial nail plate removal if pus beneath the nail (subungual abscess)**.
- **Eponychia** (proximal nail fold), **"run-around" infection** (involves both lateral folds and the eponychium).

- **Chronic paronychia: Candida albicans** (**wet work: dishwashers, bartenders, nurses, diabetics**), **inflammatory**; **avoid moisture, topical antifungals/steroids**, **eponychial marsupialisation** (crescent excision of the dorsal eponychial fold) **± nail removal** for refractory cases.
- **Differentials: herpetic whitlow**, **subungual melanoma** (persistent "paronychia"/pigment: biopsy), **squamous cell carcinoma**, **pyogenic granuloma**.

3. Felon (pulp space abscess)

- **Closed-space infection of the distal pulp**, divided into **compartments by fibrous septa** (from the skin to the periosteum of the distal phalanx): **pressure: necrosis, osteomyelitis of the distal phalanx, septic DIPJ, flexor sheath infection**.
- **Organism: *S. aureus***; **penetrating injury** (finger-prick glucose testing).
- **Throbbing pain, tense swollen pulp**.
- **Treatment: incision and drainage** with a **high lateral (mid-axial) incision** on the **non-contact side** (**ulnar side of index/middle/ring; radial side of thumb and little finger**), dividing septa, **avoiding the neurovascular bundles and the flexor sheath (stay distal to the DIP crease)**, or a **volar longitudinal incision** over the point of maximum fluctuance. **Avoid "fish-mouth" (painful scar, unstable pulp) and through-and-through incisions.**

4. Herpetic whitlow

- **Herpes simplex virus (HSV-1 or HSV-2)** infection of the fingertip; **healthcare workers (dentists, respiratory therapists), children (thumb suckers with gingivostomatitis)**.
- **Intense pain, clear vesicles (clustered), on an erythematous base**, **non-tense pulp**, prodrome of burning/tingling; **vesicles coalesce, become cloudy**.

- **Diagnosis:** clinical; **Tzanck smear (multinucleated giant cells)**, **viral PCR/culture**.
- **Treatment: non-operative**: **self-limiting (about 3 weeks)**, **protective dressing**, **oral acyclovir (if within 48-72 hours)**; **DO NOT incise** (**risk of secondary bacterial infection, systemic viraemia (encephalitis)**).

5. Pyogenic flexor tenosynovitis

- **Infection within the synovial flexor tendon sheath**: **surgical emergency** (**tendon necrosis, adhesions, rupture**, spread to deep spaces).
- **Cause: penetrating injury (commonest), haematogenous spread (*Neisseria gonorrhoeae*: disseminated gonococcal infection in sexually active young adults)**.
- **Organisms: *S. aureus*** (commonest), streptococci, mixed (**diabetics, bites**), **gonococcus**, *Mycobacterium marinum* (**fish tanks: subacute chronic tenosynovitis**: granulomas; culture at **30-32 C**; **clarithromycin/ethambutol, rifampicin**).
- **Kanavel's four cardinal signs:**
  - **Finger held in slight (semi-)flexion**,
  - **Fusiform (symmetrical) swelling** of the entire digit ("sausage digit"),
  - **Tenderness along the course of the flexor sheath**,
  - **Pain on passive extension of the finger** (**earliest and most sensitive sign**; along the sheath).
- **Sheath anatomy and spread:**
  - **Index, middle and ring finger sheaths end at the A1 pulley (MCPJ level)**.
  - **Thumb sheath (FPL) is continuous with the radial bursa**; **little finger sheath is continuous with the ulnar bursa**; the radial and ulnar bursae **communicate in about 50-80%** (**via Parona's**

space: between the pronator quadratus and FDP in the distal forearm): **"horseshoe abscess"** (thumb and little finger infection together).

- **Treatment:**
  - **Early (under about 24-48 hours) with equivocal signs**: **IV antibiotics, splinting, elevation**: **reassess within 12-24 hours**; **surgery if no improvement**.
  - **Established infection: surgical drainage**: **closed catheter irrigation (Neviaser)**: **proximal incision at the A1 pulley (palmar) and distal incision at the A5 pulley (DIP crease / mid-axial)**, **catheter irrigation of the sheath** (**limited open drainage**); **open debridement (Bruner/mid-axial incisions)** for severe/late infections with necrotic tendon.
  - **Post-op:** IV antibiotics, early ROM, elevation.
- **Complications: stiffness/adhesions (commonest)**, tendon necrosis/rupture, **amputation (diabetes, peripheral vascular disease, renal failure, subcutaneous purulence, delayed presentation)**: **Pang classification** of outcomes.

6. Deep space infections of the hand

Thenar space

- **Boundaries: radial to the oblique septum (from the third metacarpal) between the flexor tendons of the index and the adductor pollicis**; **lies deep to the index flexor tendons, superficial to the adductor pollicis**.
- **Signs: thumb held abducted** (**widened first web space**), **swelling of the thenar eminence and first web**, **pain on thumb opposition/flexion**.
- **Drainage: dual incisions: volar (thenar crease parallel, avoid the recurrent motor branch) and dorsal (first web, longitudinal)**; **avoid incisions across the first web (contracture)**.

Midpalmar space

- **Ulnar to the oblique septum**, **deep to the flexor tendons of the middle, ring and little fingers**, superficial to the interossei/metacarpals.
- **Signs: loss of palmar concavity** (convexity), **dorsal hand swelling**, painful middle and ring finger motion.
- **Drainage: palmar transverse/curved incision**.

Hypothenar space

- **Rare**; **ulnar border of the palm**; incision on the ulnar side.

Collar-button (web space) abscess

- **Infection of the interdigital web space** (**fissure or callus in labourers**): **dumbbell shape (volar and dorsal components connected through the palmar fascia)**: **abducted adjacent fingers**, **dorsal swelling**.
- **Drainage: volar (zig-zag or longitudinal) and dorsal (longitudinal) incisions**; **do not incise across the web (contracture)**.

Parona's space infection

- **Distal forearm, between pronator quadratus and FDP**: from **radial/ulnar bursa spread**; **forearm swelling, pain with finger flexion**, **acute carpal tunnel syndrome**; drainage with **carpal tunnel release**.

7. Septic arthritis of the hand and wrist

- **Penetrating trauma, bites (MCPJ: fight bite), haematogenous**; **pain on axial loading and motion**, joint swelling.
- **Aspirate** (diagnostic; small joints may be dry), **radiograph** (foreign body, bony involvement).
- **Treatment: surgical irrigation and debridement (arthrotomy/arthroscopy for wrist)**, **IV antibiotics**; **wrist septic**

arthritis: consider *Neisseria gonorrhoeae* (young sexually active), crystal disease (gout/CPPD: differential).

8. Bites

- **Human bites / fight bite (clenched-fist injury)**: see extensor tendons: **MCPJ**, ***Eikenella corrodens***, **streptococci, *S. aureus*, anaerobes (Prevotella, Fusobacterium)**; **co-amoxiclav**; **explore, irrigate, leave open**; **do not close primarily**.
- **Dog bites: *Pasteurella multocida*** (**also cats**), ***Capnocytophaga canimorsus*** (**asplenic, alcoholics: fulminant sepsis**), *S. aureus*, streptococci, anaerobes; **crush injury** component.
- **Cat bites: *Pasteurella multocida* (rapid onset, within 12-24 hours)**: **deep puncture wounds (sharp teeth) penetrate joints/sheaths: higher infection rate (about 30-50%) than dog bites**; **cat scratch disease** (***Bartonella henselae***: regional lymphadenopathy).
- **Antibiotic of choice for animal bites: amoxicillin-clavulanate (co-amoxiclav)**; **penicillin allergy: doxycycline (or ciprofloxacin/co-trimoxazole) + clindamycin/metronidazole**. ***Pasteurella* is resistant to first-generation cephalosporins (cefalexin), flucloxacillin, clindamycin and erythromycin** (alone).
- **Rabies prophylaxis** (dogs, bats, monkeys; India: high burden: **post-exposure prophylaxis: wound washing, rabies vaccine + rabies immunoglobulin (category III)**), **tetanus**.
- **Rat bites**: ***Streptobacillus moniliformis*** (**rat-bite fever**), ***Spirillum minus***.
- **Snake bite** (India: **Big Four: Indian cobra, common krait, Russell's viper, saw-scaled viper**): local tissue necrosis; **compartment syndrome is rare: fasciotomy only with measured raised pressures** (not routinely); **anti-snake venom (ASV)** (polyvalent).

9. Necrotising fasciitis

- **Rapidly spreading infection of the fascia and subcutaneous tissue**: **surgical emergency**; **mortality about 20-30%**.
- **Types: Type I (polymicrobial: commonest; diabetics, IV drug users)**; **Type II (monomicrobial: group A *Streptococcus pyogenes* ± *S. aureus*/MRSA)**: **toxic shock**; **Type III (*Vibrio vulnificus*: seawater, seafood; Clostridium: gas gangrene)**; (Type IV fungal).
- **Signs: pain out of proportion to clinical findings (earliest)**, **rapid spread**, **tense oedema beyond erythema**, **skin bullae, crepitus, grey/dusky skin, anaesthesia**, **systemic toxicity (sepsis)**.
- **LRINEC score (Laboratory Risk Indicator for Necrotising Fasciitis)**: **CRP, WBC, haemoglobin, sodium, creatinine, glucose**: **score ≥6 suspicious; ≥8 high risk** (**low sensitivity: do not rely on it to exclude**).
- **"Finger test"** at the bedside/operation: **lack of bleeding, "dishwater" fluid, fascia separates easily from tissue with finger dissection**.
- **Treatment: emergency radical surgical debridement (repeat every 24-48 hours)**, **broad-spectrum IV antibiotics (carbapenem/piperacillin-tazobactam + clindamycin (toxin suppression) + vancomycin/linezolid for MRSA)**, **ICU support**, **IV immunoglobulin (streptococcal toxic shock)**, **amputation** if needed; **hyperbaric oxygen** (adjunct, unproven).

10. Atypical and chronic infections

- ***Mycobacterium marinum*** (**"fish tank granuloma"**: aquarium workers, fishermen): **chronic tenosynovitis, nodules along lymphatics (sporotrichoid spread)**; **culture at 30-32 C (Lowenstein-Jensen)**, **biopsy (granulomas)**; **clarithromycin + ethambutol (± rifampicin) for 3-6 months**, synovectomy.

- ***M. tuberculosis***: **compound palmar ganglion (TB tenosynovitis of the flexor sheaths and radial/ulnar bursae: "rice bodies", hourglass swelling above and below the flexor retinaculum: cross-fluctuation)**; **tuberculous dactylitis (spina ventosa: children; expansile phalangeal/metacarpal lesion)**; **anti-tubercular therapy (ATT) + synovectomy**.
- **Sporotrichosis** (***Sporothrix schenckii***: **rose gardeners' disease, thorn pricks**): **nodular lymphangitis**; **itraconazole**.
- **Atypical mycobacteria** (*M. avium-intracellulare*, *M. kansasii*): immunocompromised.
- **Fungal: onychomycosis (*Trichophyton rubrum*)**: oral terbinafine.
- **Leprosy (*Mycobacterium leprae*)**: **ulnar nerve (commonest), median nerve (claw hand)**, **thickened nerves**, **neuropathic hands**; tendon transfers (**Indian experience: Brand, Fritschi**) after the disease is inactive.
- **Orf (parapoxvirus: sheep/goats)**, **milker's nodules**: self-limiting.
- **Anthrax** (rare; black eschar).
- **Gonococcal tenosynovitis/septic arthritis**: **migratory polyarthralgia, tenosynovitis (dorsal hand/wrist), pustular skin lesions** (disseminated gonococcal infection); **ceftriaxone**.

11. Special hosts

- **Diabetics:** polymicrobial (**gram-negatives, anaerobes**), atypical presentations, **higher amputation risk**, **worse outcomes in flexor tenosynovitis**; **tight glucose control**.
- **IV drug users: MRSA**, **Eikenella** (licking needles), **pseudomonas**, **necrotising fasciitis**, **injection into the dorsal hand/forearm**; **high-pressure injection injuries** (see below).
- **Immunocompromised/HIV**: atypical organisms, fungi.

12. High-pressure injection injuries

- **Paint, grease, hydraulic fluid, solvents** under high pressure (**over 7,000 kPa / about 1,000 psi**): **index finger of the non-dominant hand** (commonest).
- **Innocuous-looking pinhole entry wound** with **extensive tracking along tissue planes and sheaths**; **chemical inflammation, ischaemia, secondary infection**.
- **Surgical emergency: wide surgical debridement (decompression) within 6 hours**; **oil-based paints/solvents cause the most damage (higher amputation rates up to about 50-60%)**; **water and air** less damaging; **grease**: intermediate.
- **Amputation risk** increases with **delay (over 6-10 hours)**, **oil-based paint**, **higher pressure**, **injection into the digit (vs palm)**.
- **Radiographs** (lead-based paint visible; air), **no role for steroids (uncertain)**, broad-spectrum antibiotics.

> Exam pearl: **Kanavel signs: flexed posture, fusiform swelling, sheath tenderness, pain on passive extension (earliest/most sensitive). Thumb sheath = radial bursa; little finger = ulnar bursa; communicate via Parona's space (horseshoe abscess). Closed catheter irrigation (A1 to A5).**

> Exam pearl: **Felon: mid-axial incision on the non-contact side, divide septa; avoid fish-mouth. Herpetic whitlow: do not incise. Chronic paronychia: Candida (wet work); marsupialisation. Thenar space: thumb abducted; midpalmar: loss of palmar concavity; collar-button: dual incisions, never across the web.**

Exam pearl: **Bites: human = Eikenella; dog/cat = Pasteurella (cat bites infect more); Capnocytophaga in asplenics; all = co-amoxiclav. Necrotising fasciitis: pain out of proportion, LRINEC ≥6 (insensitive), finger test, radical debridement + clindamycin. M. marinum: fish tank, culture at 30-32 C. Compound palmar ganglion = TB with rice bodies. High-pressure injection: oil-based paint worst; emergency debridement.**

Written by Claude for Ortho preSSaleel · updated 2026-11-14. Check doubtful points in a standard textbook.